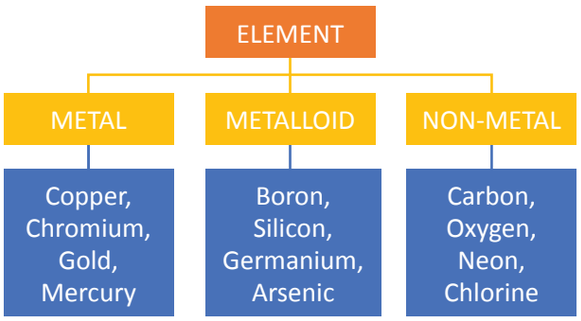


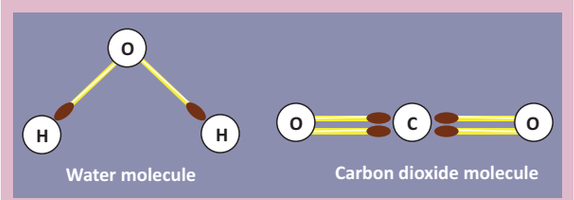
10.1.2 Compounds

When two or more elements combine chemically to form a new substance, the new substance is called a compound. For example, cane sugar is made up of three elements carbon, hydrogen and oxygen. The chemical forumula of cane sugar is $C_{12}H_{22}O_{11}$.

A compound has properties that are different from those of the elements from which it is made. Common salt, also known as sodium chloride, is a compound. It is added to give taste to our food. It is a compound made up of a metal, sodium and a non-metal, chlorine.

Activity 2

Make models of the molecules of compounds by using match sticks and clay balls as shown below,





Compounds of phosphorous, nitrogen and potassium are used in fertilizers. Silicon compounds are of immense importance in the computer industry. Compounds of fluorine are used in our toothpastes as they strengthen our teeth.

Table 10.1 Difference between elements and compounds.

Element	Compound
Made up of only one kind of atom.	Made up of more than one kind of atom.
The smallest particle that retains all its properties is an atom.	The smallest particle that retains all its properties is the molecule.
Cannot be broken down into simpler substances.	Can be broken down into elements by chemical methods.

10.1.3 Mixtures

A mixture is an impure substance. It contains *two or more kinds of elements or compounds or both physically mixed together in any ratio.*For example, tap water is a mixture of water and some dissolved salts. Lemonade is a mixture of lemon juice, sugar and water. Air is a mixture of nitrogen, oxygen, carbon dioxide, water vapour and other gases. Soil is a mixture of clay, sand and various salts. Milk, ice cream, rock salt, tea, smoke, wood, sea water, blood, tooth paste and paint are some other examples of mixtures. Alloys are mixtures of metals.



Figure 10.3 Mixtures

More to Know

LPG – Liquefied Petroleum Gas

It is highly inflammable hydrocarbon gas. It contains mixture of butane and propane gases. LPG, liquefied through pressurisation, is used for heating, cooking, auto fuel etc.



10.1.4 Differences Between Compound and Mixture

There are differences between compounds and mixtures. This can be shown by the following activity.

IX_SCI_EM_Unit-10_CHE.indd 115 06-02-2020 11:47:32